Vithlani Darsh Ashokbhai

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Education

Jain (Deemed-to-be University) 2025–2028
B.Tech CSE (AI & ML), CGPA: 8.10

Brindavan College of Engineering 2024–2025
B.Tech CSE (AI & ML), CGPA: 8.55

Experience

AI Intern – Foundations of Artificial Intelligence Apr 2025 – May 2025
Microsoft Initiative (Edunet Foundation in collaboration with AICTE)

- Developed a banking FAQ chatbot using TF-IDF vectorization and Logistic Regression for intent classification.
- Built preprocessing and inference pipelines using Python and Scikit-learn for real-time query prediction.

Student Intern – OmniForge Apr 2026 – Present

- Created and annotated datasets for document understanding and computer vision workflows involving medical documents.
- Generated bounding boxes, segmentation masks, and coordinate annotations for supervised machine learning pipelines.

Projects

Multimodal Visual Search Engine | Python, CLIP, FAISS, ONNX Runtime, FastAPI, Docker 2026
GitHub Live Demo

- Built a multimodal semantic search engine supporting text, image, and voice retrieval across **10,000+ images**.
- Optimized CLIP inference via ONNX INT8 quantization, reducing model size from **350MB to 90MB**; implemented FAISS vector retrieval.
- Designed and deployed a Dockerized microservice architecture on Hugging Face Spaces; published a 10,000+ image dataset on Kaggle and Hugging Face.

NARA AI-Powered Personalized Nutrition & Food Recommendation Platform | Python, FastAPI, PostgreSQL, Redis, Kafka, Docker, LightGBM, XGBoost, PyTorch 2026
GitHub

- Engineered a **19M+ record synthetic behavioral dataset** across 10 tables (5K users, 4.6M meal logs, 7.3M interactions, 2.1M reorder events) using NFHS-5, Census 2011, and ICMR distributions; validated quality through a **151-check framework** ensuring biological, cultural, and cross-table consistency.
- Designed and trained **15 production-grade ML models** across 5 systems: recommendation ranking (Logistic, XGBoost, LightGBM), cold-start personalization (KNN, MLP, Wide & Deep), health-compliance scoring (Rules, Random Forest, XGBoost+SHAP), meal-occasion classification (Decision Tree, Random Forest, XGBoost), and reorder prediction (Logistic, Random Forest). Achieved **NDCG@10 ≈ 0.47**, **76.8% cold-start accuracy**, **99.6% Top-3 accuracy**, **97.4% meal-occasion F1**, **AUC 0.988** for health compliance, and **AUC 0.745** for reorder prediction.
- Architected distributed ML platform using Kafka, Redis, PostgreSQL, FastAPI, and Docker; leveraged SHAP explainability to identify key drivers (GI score, stress level, carbohydrate intake) aligned with ADA 2023 guidelines, demonstrating model generalization.

Nourish Food & Health Vision-Language Model | Python, PyTorch, Transformers, ViT, FastAPI, ChromaDB, SQLite 2026
GitHub

- Engineered a **domain-specific Vision-Language Model (VLM)** for food understanding and health-aware question answering from images and natural language queries.
- Designed a **multimodal transformer architecture** combining Vision Transformers (ViT), cross-attention fusion, and transformer decoders for vision-language reasoning.
- Trained the model on **500K+ nutrition and medical text samples** and **75K+ food images** using staged pretraining and multimodal fine-tuning.
- Achieved **77% Top-1 accuracy on Food-101** and deployed a FastAPI inference service with **ChromaDB retrieval** and SQLite-based user profile storage.

Technical Skills

Languages: Python, C++
ML/AI: PyTorch, Scikit-learn, CLIP, FAISS, Transformers, Computer Vision, NLP
Backend: FastAPI, PostgreSQL, Redis, Kafka, REST APIs
DevOps & Tools: Docker, Nginx, ONNX Runtime, Git, Linux

Leadership & Activities

Technical Member – Turing Club Feb 2026 – Present
Built and operated EmojiLang, an emoji-based programming language and coding judge used by 15+ teams during a live coding competition.

Open Source Contributions

- Released the **NARA Synthetic Recommendation Dataset**, a **19M+ record** benchmark spanning 5K users, 4.6M meal logs, 7.3M recommendation interactions, and 2.1M reorder events.
- Curated and published a **10K+ image dataset** for computer vision and multimodal search applications.
- Opened a pull request for **Qdrant FastEmbed** proposing fixes for image preprocessing and custom model resolution bugs, including regression tests and reproducible bug reports.